

**Multiple Choice:**

1. Transient disturbance is produced in a circuit whenever
 - A. it is suddenly connected or disconnected from the supply
 - B. it is shorted
 - C. its applied voltage is changed suddenly
 - D. all of the above
2. To double the resonant frequency of an LC circuit with a fixed value of L, the capacitance, C, must be
 - A. doubled.
 - B. quadrupled.
 - C. reduced by one-half.
 - D. reduced by one-quarter.
3. After two-time constants, the capacitor in an RC circuit is charged to what percentage of the supply voltage?
 - A. 36.8 %
 - B. 86.5 %
 - C. 63.2 %
 - D. 95 %
4. In an R-L circuit connected to an alternating sinusoidal voltage, size of transient current primarily depends on
 - A. the instant in the voltage cycle at which circuit is closed
 - B. the peak value of steady-state current
 - C. the circuit impedance
 - D. the voltage frequency
5. The resonant frequency of an LC circuit is the frequency where
 - A. $X_L = 0 \Omega$ and $X_C = 0 \Omega$.
 - B. $X_L = X_C$
 - C. X_L and r_s of the coil are equal.
 - D. X_L and X_C are in phase.
6. An uncharged capacitor of $0.2 \mu\text{F}$ is connected to a 100 V, d.c. supply through a resistor of 100 k Ω . Determine the capacitor voltage 10 ms after the voltage has been applied.
 - A. 39.35 V
 - B. 35.39 V
 - C. 33.95 V
 - D. 39.53 V
7. What is the term for the time required for the capacitor in an RC circuit to be charged to 63.2% of the supply voltage?
 - A. An exponential rate of one.
 - B. One time constant.
 - C. One exponential period.
 - D. A time factor of one.
8. A higher Q for a resonant circuit provides a
 - A. dampened response curve.
 - B. wider bandwidth.
 - C. narrower bandwidth.
 - D. None of the above.
9. In a low Q parallel resonant circuit, when $X_L = X_C$.
 - A. $I_L = I_C$.
 - B. I_L is less than I_C .
 - C. I_C is less than I_L .
 - D. I_L is more than I_C .
10. What will happen to a parallel ac-circuit if its line frequency is more than the resonant frequency?
 - A. Becomes purely resistive
 - B. Becomes purely capacitive
 - C. Becomes inductive
 - D. Becomes capacitive
11. Below resonance, a series LC circuit appears
 - A. inductive.
 - B. resistive.
 - C. capacitive.
 - D. None of the above
12. A circuit consists of an uncharged capacitor connected in series with a 50 k Ω resistor and has a time constant of 15 ms. Determine the capacitance of the capacitor.
 - A. 0.309 μF
 - B. 3.09 μF
 - C. 0.039 μF
 - D. 309 μF
13. Which of the following can provide a higher Q?
 - A. a higher L/C ratio.
 - B. a lower L/C ratio.
 - C. more resistance in series with the coil.
 - D. Either B or C.
14. The phase angle of an LC circuit at resonance is
 - A. 0° .
 - B. 180° .
 - C. $+90^\circ$.
 - D. -90° .
15. Characteristics of the current in a series R-L-C circuit at resonance.
 - A. It is dc
 - B. It is a minimum
 - C. It is zero
 - D. It is at maximum
16. A parallel circuit at resonance would mean, the circuit is
 - A. resistive
 - B. inductive
 - C. reactive
 - D. capacitive
17. Double-energy transients are produced in circuits consisting of
 - A. two or more resistors
 - B. resistance and inductance
 - C. resistance and capacitance
 - D. resistance, inductance and capacitance
18. The impedance of a series LC circuit at resonance is
 - A. maximum.
 - B. nearly infinite.
 - C. minimum.
 - D. Both A and B.
19. A coil and capacitor are in series. The inductance is 88 mH and the capacitance is 1000 pF. What is the resonant frequency?
 - A. 17 kHz
 - B. 17 MHz
 - C. 540 Hz
 - D. 540 kHz



20. A series LC circuit has a Q of 100 at resonance. If $V_{in} = 5 \text{ mV p-p}$, how much is the voltage across C?

- A. $50 \mu\text{V p-p}$.
- B. 5 mV p-p .
- C. 50 mV p-p .
- D. 500 mV p-p .

21. There are no transients in pure resistive circuits because they

- A. offer high resistance
- B. obey Ohm's law
- C. have no stored energy
- D. are linear circuits

22. Above resonance, a parallel LC circuit appears

- A. inductive.
- B. resistive.
- C. capacitive.
- D. None of the above.

23. What is the meaning of the term "time constant of an RL circuit"? The time required for the:

- A. Current in the circuit to build up to 36.8% of the maximum value.
- B. Voltage in the circuit to build up to 63.2% of the maximum value.
- C. Current in the circuit to build up to 63.2% of the maximum value.
- D. Voltage in the circuit to build up to 36.8% of the maximum value.

24. A $10 \mu\text{F}$ capacitor is charged to 120 V and then discharged through a $1.5 \text{ M}\Omega$ resistor. Find how long it takes for the voltage to fall to 25 V.

- A. 25.33 s
- B. 22.53 s
- C. 23.53 s
- D. 21.53 s

25. The transient current in a loss-free L-C circuit when excited from an ac source is a/an _____ sine wave.

- A. Overdamped
- B. Undamped
- C. Under damped
- D. Critically damped

26. When either L or C is increased, the resonant frequency of an LC circuit

- A. decrease.
- B. increases.
- C. doesn't change.
- D. it cannot be determined.

27. The current at the resonant frequency of a series LC circuit is 10 mA p-p . What is the value of current at the half-power points?

- A. 7.07 mA p-p
- B. 14.14 mA p-p .
- C. 5 mA p-p .
- D. 10 mA p-p .

28. Transient currents in electrical circuit are associated with

- A. Inductors
- B. Capacitors
- C. Resistors
- D. Both A and B

29. A series circuit at resonance would mean, the circuit is

- A. resistive
- B. capacitive
- C. inductive
- D. reactive

30. The impedance of a parallel LC circuit at resonance is

- A. zero.
- B. maximum.
- C. minimum.
- D. equal to the r_s of the coil.

31. In a series ac-circuit, if the line frequency is more than the resonant frequency, the circuit behaves as

- A. resistive
- B. reactive
- C. inductive
- D. capacitive

32. A capacitor is connected in series with a voltmeter of resistance $750 \text{ k}\Omega$ and a battery. When the voltmeter reading is steady the battery is replaced with a shorting link. If it takes 17s for the voltmeter reading to fall to two-thirds of its original value, determine the capacitance of the capacitor.

- A. $55.90 \mu\text{F}$
- B. $55.80 \mu\text{F}$
- C. $55.10 \mu\text{F}$
- D. $55.60 \mu\text{F}$

33. What is the meaning of the term "time constant of an RC circuit"? The time required to charge the capacitor in the circuit to:

- A. 23.7% of the supply voltage.
- B. 36.8% of the supply voltage.
- C. 57.3% of the supply voltage.
- D. 63.2% of the supply voltage.

34. The total line current, I_T , of a parallel LC circuit at resonance is

- A. minimum.
- B. maximum.
- C. equal to I_L and I_C .
- D. Q times larger than I_L or I_C .

35. A series-resonant circuit is to be made for 14.1 MHz. A coil of $13.5 \mu\text{H}$ is available. What capacitor size is needed?

- A. $0.945 \mu\text{F}$
- B. 94.5 pF
- C. 9.45 pF
- D. 945 pF

36. After two-time constants, the capacitor in an RC circuit is discharged to what percentage of the starting voltage?

- A. 86.5 %
- B. 63.2 %
- C. 13.5 %
- D. 36.8 %

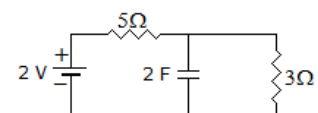
37. A capacitor of 470 pF is in parallel with an inductor of $4.44 \mu\text{H}$. What is the resonant frequency?

- A. 3.49 MHz
- B. 13.0 MHz
- C. 3.49 kHz
- D. 13.0 GHz

38. If an ac-series circuit is supplied with a source whose frequency is less than that of f_r , the circuit becomes

- A. resistive
- B. reactive
- C. inductive
- D. capacitive

39. The time constant of the circuit in figure is





- A. 6 sec C. 10 sec
B. 15/4 sec D. 16 sec
40. An inductor and a capacitor are connected in parallel. At the resonant frequency the resulting impedance is:
- A. maximum
B. minimum
C. totally reactive
D. totally inductive
41. A "high Q" resonant circuit is one which:
- A. carries a high quiescent current
B. is highly selective
C. has a wide bandwidth
D. uses a high value inductance
42. What is the cause of a minimum Q on a single-tuned LC circuit?
- A. Decreased series resistor
B. Decreased shunt resistor
C. Increased shunt resistor
D. Decreased capacitance
43. The Q of a parallel resonant circuit can be lowered by
- A. placing resistor in parallel with the tank.
B. adding more resistance in series with the coil.
C. decreasing the value of L or C
D. Both A and B.
44. What is the term for the time required for the current in an RL circuit to build up to 63.2% of the maximum value?
- A. One time constant.
B. An exponential period of one.
C. A time factor of one.
D. One exponential rate.
45. The current at resonance in a series LC circuit is
- A. zero.
B. minimum.
C. different in each component.
D. maximum.
46. An inductor and a capacitor form a resonant circuit. The capacitor value is increased by four times. The resonant frequency will:
- A. increase by four times
B. double
C. decrease to half
D. decrease to one quarter
47. The ability of an LC circuit to supply complete sine waves when the input to the tank is only a pulse is called
- A. tuning.
B. the flywheel effect.
C. antiresonance.
D. its Q.
48. An inductor and a capacitor form a resonant circuit. If the value of the inductor is decreased by a factor of four, the resonant frequency will:
- A. increase by a factor of four
B. increase by a factor of two
C. decrease by a factor of two
D. decrease by a factor of four
49. An inductor and a capacitor are connected in series. At the resonant frequency the resulting impedance is:
- A. maximum
B. minimum
C. totally reactive
D. totally inductive
50. The transients which are produced due to sudden but energetic changes from one steady state of a circuit to another are called _____ transients.
- A. initiation
B. transition
C. relaxation
D. subsidence

END